Yenepoya University
Project – Low Level Design
on
End-to-End Encryption for Manufacturing Cloud Apps

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1. Introduction

1.1 Scope of the Document

This document provides the Low-Level Design (LLD) of the project "Secure End-to-End Encryption for Retail Cloud Applications". It explains the internal working of the system, including modules, functions, data handling, and detailed process flow used for secure communication in retail cloud environments.

1.2 Intended Audience

This document is intended for BCA and BSc Computer Science students, developers, project evaluators, and academic supervisors who need a detailed understanding of the system design and implementation.

1.3 System Overview

The system is a web-based application that ensures secure communication in retail cloud environments by encrypting and decrypting messages using the Fernet encryption technique. It simulates secure data transmission for real-world retail use cases such as protecting customer information, securing billing communications, and safeguarding payment transactions.

2. System Design

2.1 Application Design

The application follows a client-server model where the user interacts with a web page and the backend processes the data

- The frontend is created using HTML and CSS. It allows the retail user to enter messages and view the output.
- The backend is developed using Python Flask, which handles user requests.
- The encryption process uses the Fernet algorithm to secure the retail message.

2.2 Process Flow

1. The Sender (retail staff) enters a message in the application.
2. The message is encrypted using a secret key (Fernet algorithm).
3. The encrypted message is stored or transmitted securely.
4. The Receiver enters the same secret key to access the message.
5. The system decrypts the message and displays the original message.

2.3 Information Flow

The information flow in the system describes how data is processed and transformed from input to output.

- **Input:** The sender enters a plain text message into the system.
- **Processing:** The message is encrypted into ciphertext and later decrypted using the Fernet algorithm and a secret key.
- **Output:** The system produces an encrypted message for secure storage and the original message after decryption.

The system ensures that sensitive data is never stored in plain text form, thereby maintaining data confidentiality and security.

2.4 Components Design

The system is divided into different components, each performing a specific function. This modular design makes the system simple and easy to manage.

1. User Interface Component

- Developed using HTML and CSS
- Allows the retail sender to enter messages
- Displays encrypted and decrypted output to the receiver

2. Backend Component (Flask)

- Developed using Python Flask
- Handles user requests and responses
- Connects frontend with encryption and decryption modules

3. Encryption Component

Encrypts the plain text retail message

Uses Fernet algorithm with a secret key

Converts readable text into encrypted format

4. Decryption Component

Decrypts the encrypted message

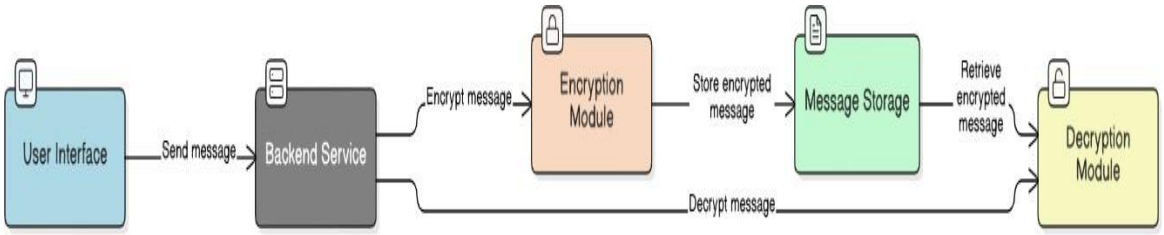
Uses the same secret key

Converts encrypted data back to original message

5. Storage Component

Stores encrypted messages in a file (messages.txt)

Ensures that only encrypted data is saved



4. Interfaces

4.1 User Interface

The user interface is built using HTML and CSS. It includes:

A text box to enter the message

An **Encrypt** button

A **Decrypt** button

A section to display the result

The interface is simple and easy to use.

4.2 Backend Interface

The backend is developed using Flask. It has two main routes:

/ encrpt→ Encrypts the message

/decrypt → Decrypts the message

These routes connect the frontend with the backend and handle data processing.

3. Data Design

3.1 Data Model

The system uses a simple data model to handle secure communication:

Plain Text Message: Input provided by the retail sender

Encrypted Message: Output generated after encryption

Secret Key: Used for both encryption and decryption

The data is converted from readable format to encrypted format to ensure security.

3.2 Data Storage

The system stores data using files:

messages.txt → Stores encrypted retail messages

key.key → Stores the secret encryption key

Only encrypted data is stored to maintain confidentiality and prevent unauthorized access.

3.3 Data Access Mechanism

The system accesses data using file handling operations:

The key is read from the key.key file

Encrypted data is written to and read from messages.txt

The same key is used for both encryption and decryption

This ensures secure and efficient data handling.

5. State and Session Management

The system uses role-based authentication to control access.

Login & Authentication

Users must log in with valid credentials. The system supports three roles: Admin, Cashier, and Customer Support. Access is restricted based on role permissions.

Role-Based Access Control

Admin: Full access (including audit logs and key management)

Cashier: Access limited to billing and transaction messages at assigned till or terminal

Customer Support: Access limited to customer communication messages; no access to billing or financial data

Session Handling

Session data is stored in browser memory (localStorage) and cleared on logout.

Audit Trail

Key actions (login, logout, message activity, failed login attempts) are logged with timestamps. Audit logs are accessible only to Admins.

6. Non-Functional Requirements

6.1 Security Aspects

Implements End-to-End Encryption (E2EE)

Ensures confidentiality of messages

Prevents storage of plain text data

Uses a secret key for encryption and decryption

Provides role-based access control for secure operations

6.2 Performance Aspects

Fast execution of encryption and decryption operations

Minimal response time for user actions

Efficient handling of data storage and retrieval

6.3 Usability

Simple and user-friendly interface

Easy navigation for different user roles

Clear display of input and output messages

7. References

Python Official Documentation
Flask Framework Documentation
Cryptography Library (Fernet) Documentation